Chapter 10
Traffic Shaping and Flow Management in Wide Area Networks

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ABSTRACT

Wide area networks, which are the basis of access to information and communication today, form the backbone of the internet. The ever-increasing data traffic, different application types, and service quality requirements of different systems used in local area networks cause problems such as congestion and loss of performance in networks due to the limited bandwidth received from the internet service provider. This study focuses on the effects of QoS functions and critical network management tools in wide area networks with limited bandwidth, such as traffic shaping, traffic control, bandwidth management, and data traffic prioritization methods, on network performance. The results obtained from this study show that traffic shaping methods are used according to the need to adapt to the limited bandwidth within the framework of the designed rules and are important in improving the performance of the managed network and controlling sudden increases and traffic fluctuations.

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INTRODUCTION

Nowadays, network traffic density is increasing day by day. It is of great importance to transfer information flow uninterruptedly and securely in wide area networks. The increase in data amounts negatively affects the operating performance and stability of wide area networks with limited bandwidth. Frequent network traffic congestion or delays in network traffic due to intense data flow both reduce efficiency and interrupt critical information flow.

In wide area networks, it is difficult to manage network traffic smoothly due to hardware costs, balancing network load and ensuring network performance. Due to the increase in the number of users, applications, different systems and network devices, effective management of the limited bandwidth received from the internet service provider has become a necessity. It is important to find out solutions to how to use traffic shaping and flow management methods in wide area networks.

Network communication quality of service (QoS) is a technology used to manage traffic in a network and prioritize planned data types. Traffic shaping and flow management methods within the scope of QoS aim to solve these problems. When QoS methods are not applied to network devices, serious disruptions can occur in data traffic that needs to be transmitted, especially in emergency situations. However, faced with increasing Internet traffic and diverse QoS requirements, traditional centralized network architectures encounter challenges (She et al., 2025). The Internet best-effort service led to the lacking deployment of network QoS on the Internet architecture since there are problems associated with currently used short path routing (Elbasheer et al., 2021).

In this study the main purpose is to limit the bandwidth of existing systems used in local area networks so that they do not exceed the limited capacity defined by the Internet service provider when going to the wide area network and to determine the important network traffic and to ensure uninterrupted wide area network connection. This way, the replies will be given to the following questions.

- Is managing the network by traffic shaping in wide area networks with limited bandwidth sufficient for network performance?
- Which method is useful in using the QoS model to prioritize critical traffic, optimize network performance, create traffic classes, and minimize data loss?

In the application phase of this study, a network structure consisting of a Cisco 4451 router, a Cisco 9300 network switch, and a group of computers was designed. One of the computers was used to carry out performance and bandwidth tests for traffic coming from other systems in the network. Limiting bandwidth was not the only purpose of the study. Another purpose was to determine the important network

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